THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the equation $2x + 6y = 34$, when $x = 2$ the value of y is: 1

(a) 6 **(b)** 4 **(c)** 7 **(d)** 5

2. The total surface area of a cube of edge 4 cm is: 1

(a) 106 cm^2 **(b)** 96 cm^2 **(c)** 86 cm^2 **(d)** 117 cm^2

3. The distance of the point $(3, 4)$ from the origin is: 1

(a) 7 **(b)** 4 **(c)** 5 **(d)** 6

4. The y-intercept of the line $y = 2x + 5$ is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

5. The decimal expansion of $\frac{1}{3}$ is: 1

(a) $0.0\overline{3}$ **(b)** $3.\overline{3}$ **(c)** $0.\overline{6}$ **(d)** $0.\overline{3}$

6. Using $(a + b)^2$, the value of 31^2 is: 1

(a) 961 **(b)** 962 **(c)** 960 **(d)** 963

7. The angle subtended by an arc at the centre is $50°$. The angle it subtends at a point on the remaining part of the circle is: 1

(a) $15°$ **(b)** $45°$ **(c)** $25°$ **(d)** $35°$

8. The area of a triangle with base 12 cm and height 8 cm is (cm^2): 1

(a) 50 **(b)** 49 **(c)** 47 **(d)** 48

9. A ticket is drawn from tickets numbered 1 to 10. The probability that the number is a multiple of 2 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{2}{3}$

10. The next term of the GP $1, 2, 4, 8, \ldots$ is: 1

(a) 15 **(b)** 17 **(c)** 18 **(d)** 16

11. "Every square is a rectangle" can be written in if–then form as: 1

(a) If it is a rectangle, then a figure is a square.

(b) A figure is a square or it is a rectangle.

(c) If a figure is a square, then it is a rectangle.

(d) If not (a figure is a square), then it is a rectangle.

12. A stacked bar for a family's monthly food use shows rice 32 kg, wheat 24 kg, pulses 39 kg. The total length of the bar is (in kg): 1

(a) 96 **(b)** 95 **(c)** 94 **(d)** 97

13. The greatest of the common divisors of 30 and 22, found from their divisor lists, is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

14. Which quadrilateral always has all four sides equal? 1

(a) parallelogram **(b)** rhombus **(c)** rectangle **(d)** kite

15. The graph of a linear equation in two variables is: 1

(a) a pair of points **(b)** a circle **(c)** a straight line **(d)** a parabola

16. The lateral surface area of a cuboid 9 cm $\times$ 5 cm $\times$ 3 cm (height 3 cm) is: 1

(a) 75 cm^2 **(b)** 93 cm^2 **(c)** 102 cm^2 **(d)** 84 cm^2

17. The midpoint of the segment joining $(2, 4)$ and $(6, 8)$ is: 1

(a) $(-4, 6)$ **(b)** $(6, 4)$ **(c)** $(4, -6)$ **(d)** $(4, 6)$

18. The slope of the line $y = 3x + 2$ is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 5

19. A decimal expansion that ends after a finite number of digits is called: 1

(a) recurring **(b)** irrational
(c) non-terminating and non-repeating **(d)** terminating

20. Using $a^2 - b^2$, the value of 49×51 is: 1

(a) 2500 **(b)** 2499 **(c)** 2498 **(d)** 2501

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Write $4x + 8y = 5$ in the form $ax + by + c = 0$ and state a, b and c. 2

22. Find the volume, total surface area and lateral surface area of a cuboid of length 9 cm, breadth 6 cm and height 3 cm. 2

23. Find the distance of the point $(6, 8)$ from the origin. 2

24. Find the zero of the polynomial $3x - 15$. 2

25. Is $\sqrt{16}$ rational or irrational? Give a reason. 2

26. Expand $(x + 8)(x - 8)$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Express $\frac{5}{12}$ as a decimal and say whether it terminates or repeats. 3

28. Factorise $x^2 + 6x + 9$. 3

29. Points A and B lie on a circle with centre O and $\angle AOB = 48^\circ$. A point C lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point D on the major arc. 3

30. Find the area of a triangle with sides 3 cm, 4 cm and 5 cm using Heron's formula. 3

31. A die was thrown 120 times and the number 4 appeared 22 times. Find the experimental probability of getting 4, and of not getting 4. 3

OR

Find the sum of the first 20 natural numbers. 3

32. Show that the statement "If $x^2 > 9$, then $x > 3$" is false by giving a counterexample. 3

OR

A girl cycles for 3 weeks of equal length. Her weekly averages are 56 min, 26 min, 41 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. AB is a diameter of a circle and C, D lie on the circle on opposite sides of AB. If $\angle CAB = 25^\circ$ and $\angle DAB = 30^\circ$, find $\angle ABC$ and $\angle ABD$. 5

34. A circular park has radius 14 m. Find its circumference and area (take $\pi = \frac{22}{7}$), and the cost of fencing it at ₹15 per metre. 5

35. A fair die is rolled once. Find 5
(i) $P(\text{an even number})$,
(ii) $P(\text{a number greater than } 4)$ and
(iii) $P(\text{an even number or a number greater than } 4)$ using the favourable outcomes.

36. An AP has first term 2 and common difference 3. Find the sum of its first 10 terms. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 6 and 7. 5

37. 20 mL of a 40% sugar solution is mixed with 60 mL of a 28% sugar solution. Find the percentage concentration of the mixture. 5

OR

Find gcd(50, 75) by listing the divisors of both numbers. 5

38. The diagonals of a rhombus are 12 cm and 16 cm. Find its side, perimeter and area. 5

OR

A taxi charges ₹31 fixed and ₹11 per km. Write the equation relating the fare y (in ₹) and the distance x km and find the fare for 4 km. 5

END OF QUESTION PAPER — Total: 80 Marks

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